

Team Members

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```
import java.util. ArrayList;
// class for Security System
class SecuritySystem {
    public void powerOn() {
        System.out.println("Security System is active.");
    }
}
```

// class for Smart Appliances

```
class SmartAppliance {
    private final String name;
```

// 1-arg constructor

```
public SmartAppliance(String name) {
    this.name = name;
}
```

```
public void powerOn() {
    System.out.println(name + " is running.");
}
```

// class for Smart Home

```
class SmartHome {
```

```
    private SecuritySystem sSystem;
```

```
    private final ArrayList<SmartAppliance> appliances;
```


// no-arg constructor

```
public SmartHome() {
```

// instantiating new security system

```
-this.system = new SecuritySystem();
```

```
-this.appliances = new ArrayList<>();
```

```
}
```

```
public void addAppliance(SmartAppliance appliance) {  
    appliances.add(appliance); // add appliance.
```

```
} // method for adding appliances to home.
```

```
public void removeAppliance(SmartAppliance appliance) {  
    appliances.remove(appliance);
```

```
} // method for removing appliances from home.
```

```
}
```

// Main class

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        SmartAppliance myCoffeeMaker = new
```

```
            SmartAppliance("Espresso Machine");
```

```
        SmartHome home = new SmartHome();
```

```
        // adding Espresso Machine (myCoffeeMaker) to home
```

```
        home.addAppliance(myCoffeeMaker);
```

```
        myCoffeeMaker.powerOn();
```

```
        home = null; // putting home = null i.e. deleting it.
```

```
        // turning coffee maker on again
```

```
        myCoffeeMaker.powerOn();
```

```
}
```


Q & A:

Q: Does the coffee maker still exist in memory?

A: Yes, it exists independently from SmartHome due to Aggregation.

Q: Can you still access the SecuritySystem?

No, since the SecuritySystem was an object inside the home and is destroyed after due to Composition.